15. September 2026: Progress Review and Next-Month Work Plan

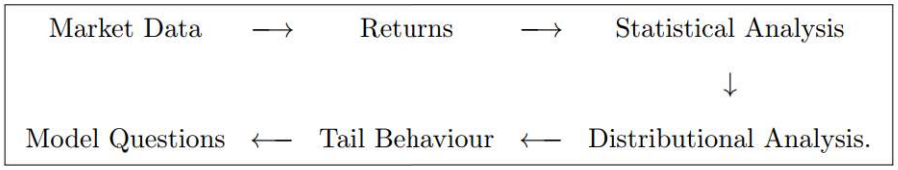
15.1 Progress Achieved So Far

The first stage of the project has been completed successfully using approximately ten years of NIFTY 50 market data.

The work completed so far includes

- acquisition and validation of historical price data;
- computation of simple and logarithmic returns;
- numerical verification of the relationship between simple and log returns;
- verification of the time-additivity property of logarithmic returns;
- estimation of mean, variance, volatility, skewness and kurtosis;
- investigation of the difference between arithmetic and logarithmic mean returns;
- statistical testing of the normality assumption;
- investigation of extreme-return frequencies;
- graphical analysis of price and return behaviour; and
- establishment of a reproducible Python-based computational workflow.

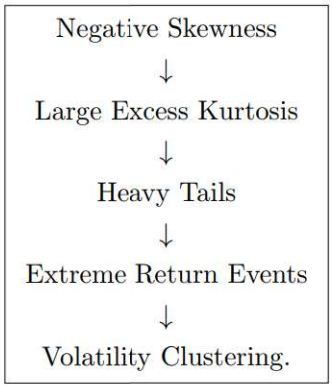
The work has therefore progressed from



15.2 Important Findings from the Present Work

The initial analysis has already produced several observations that are important for the later Black–Scholes and machine-learning stages.

The empirical return distribution exhibits



These observations are particularly important because the next stage introduces geometric Brownian motion and the classical Black–Scholes framework.

The September work will therefore not treat GBM merely as a simulation exercise.

The objective will be to compare

Behaviour Assumed by the Model	with	Behaviour Observed in Market Data.

15.3 September Objective

The main objective for September is to complete the mathematical and computational foundation of the classical option-pricing framework.

The progression for the month will be

Brownian Motion	$\longrightarrow$	GBM	$\longrightarrow$	Itô's Lemma
				$\downarrow$
Numerical Validation	$\longleftarrow$	BS Pricing	$\longleftarrow$	BS PDE.

The month will conclude with a validated computational framework for

$$C, \qquad P, \qquad \Delta, \qquad \Gamma, \qquad \nu, \qquad \sigma_{\mathrm{IV}}.$$

15.4 Week 1: Brownian Motion and Geometric Brownian Motion

15.4.1 Learning Objectives

The first objective is to understand the stochastic process underlying the Black–Scholes model.

The Brownian-motion increment satisfies

$$\Delta W = W_{t+\Delta t} - W_t$$

with

$$\Delta W \sim N(0, \Delta t).$$

Equivalently,

$$\boxed{\Delta W = \sqrt{\Delta t}\, Z, \qquad Z \sim N(0,1).}$$

The corresponding asset-price model is

$$\boxed{\frac{dS_t}{S_t} = \mu\, dt + \sigma\, dW_t.}$$

15.4.2 Computational Work

The following numerical experiments are to be completed:

1. Generate standard normal random variables.
2. Generate Brownian increments.
3. Construct complete Brownian-motion paths.
4. Verify numerically that

$$E[\Delta W] \approx 0.$$

5. Verify

$$\mathrm{Var}(\Delta W) \approx \Delta t.$$

6. Repeat the experiment for different values of Δt.
7. Demonstrate numerically that

$$\mathrm{SD}(\Delta W) \propto \sqrt{\Delta t}.$$

15.4.3 GBM Simulation

The analytical GBM solution is

$$S_t = S_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right].$$

Multiple paths are to be generated for different combinations of

$$\mu \qquad \text{and} \qquad \sigma.$$

Particular attention should be given to understanding the different roles of

$$\boxed{\mu = \text{drift}}$$

and

$$\boxed{\sigma = \text{volatility}.}$$

15.4.4 Connection with the Previous NIFTY Analysis

The empirical volatility obtained from the previous analysis provides a useful starting value,

$$\sigma \approx 0.167.$$

GBM paths should therefore be simulated using approximately the same annualized volatility as the empirical NIFTY 50 data.

15.4.5 Model-versus-Data Experiment

This is an important September experiment.

Generate a GBM return sample having approximately the same

$$\mu \qquad \text{and} \qquad \sigma$$

as the empirical NIFTY 50 sample.

Then compare

NIFTY Returns	versus	GBM Returns.

Compare

mean, standard deviation, skewness, kurtosis,

together with

histograms, QQ plots, tail frequencies.

The principal question is

> If GBM is calibrated to the same mean and volatility,
> does it reproduce the distributional behaviour
> observed in the actual market data?

15.4.6 Week 1 Expected Output

`week02_gbm_simulation.ipynb`

Brownian-Motion Plots
GBM Path Plots
Empirical-versus-GBM Comparison
Short Interpretation of Model Limitations.

15.5 Week 2: Itô's Lemma and Black–Scholes Equation

15.5.1 Learning Objectives

The objective is to understand how the stochastic asset-price model

$$\frac{dS}{S} = \mu\, dt + \sigma\, dW$$

leads to the Black–Scholes partial differential equation.

For

$$V = V(S, t),$$

Itô's lemma gives

$$dV = \left[V_t + \mu S V_S + \frac{1}{2}\sigma^2 S^2 V_{SS}\right] dt + \sigma S V_S \, dW.$$

15.5.2 Delta-Hedging Argument

Construct

$$\Pi = V - \Delta S$$

with

$$\Delta = V_S.$$

The stochastic component is then eliminated.

Using the no-arbitrage condition

$$d\Pi = r\Pi \, dt,$$

derive

$$\boxed{V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0.}$$

15.5.3 Key Questions

The following points must be understood clearly:

1. Why is Itô's lemma required instead of ordinary calculus?
2. Where does

   $$\frac{1}{2}\sigma^2 S^2 V_{SS}$$

   come from?
3. How does delta hedging eliminate the dW term?
4. Why does the physical drift

   $$\mu$$

   disappear from the final Black–Scholes PDE?
5. Why does the risk-free rate

   $$r$$

   appear in its place?

15.5.4 Analytical Pricing

Implement

$$C_{\mathrm{BS}} = SN(d_1) - Ke^{-rT}N(d_2),$$

where

$$d_1 = \frac{\ln(S/K) + \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}$$

and

$$d_2 = d_1 - \sigma\sqrt{T}.$$

The corresponding put-price formula should also be implemented.

15.5.5 Basic Validation

Verify put–call parity,

$$\boxed{C - P = S - Ke^{-rT}.}$$

Perform the test over several combinations of

$$S, \qquad K, \qquad T, \qquad r, \qquad \sigma.$$

15.5.6 Week 2 Expected Output

`black_scholes.py`
Black–Scholes Derivation Note
Call and Put Price Tables
Put–Call Parity Validation.

15.6 Week 3: Greeks and Implied Volatility

15.6.1 Greeks

Implement and understand

$$\Delta = \frac{\partial V}{\partial S},$$

$$\Gamma = \frac{\partial^2 V}{\partial S^2},$$

and

$$\nu = \frac{\partial V}{\partial \sigma}.$$

The analytical values should be independently checked using finite-difference approximations.

15.6.2 Sensitivity Analysis

Generate

$$C(S), \qquad \Delta(S), \qquad \Gamma(S), \qquad \nu(S)$$

for several maturities.

The behaviour should be examined separately in the

$$\boxed{\text{ITM}, \qquad \text{ATM}, \qquad \text{OTM}}$$

regions.

The objective is not merely to reproduce Greek formulas but to understand their financial meaning.

15.6.3 Implied Volatility

For an observed option price,

$$C_{\text{market}},$$

implied volatility satisfies

$$\boxed{C_{\text{BS}}\left(S, K, T, r, \sigma_{\text{IV}}\right) = C_{\text{market}}.}$$

Implement

$$\boxed{\text{Newton–Raphson} + \text{Bisection Fallback.}}$$

15.6.4 Recovery Test

Select a known volatility

$$\sigma_{\text{true}}$$

and generate

$$C_{\text{BS}}\left(\sigma_{\text{true}}\right).$$

Using only the generated option price, recover the volatility numerically.

The required result is

$$\boxed{\sigma_{\mathrm{IV}} \approx \sigma_{\mathrm{true}}.}$$

This test should be repeated over different

$$S/K, \qquad T, \qquad \sigma.$$

15.6.5 Week 3 Expected Output

`greeks.py`
`implied_volatility.py`
Delta, Gamma and Vega Plots
Analytical-versus-Numerical Greek Comparison
Implied-Volatility Recovery Study.

15.7 Week 4: Independent Numerical Validation

The final September stage is intended to establish confidence in the analytical Black–Scholes implementation.

Two independent numerical methods will be used:

Binomial Method	+	Monte Carlo Method.

15.7.1 Binomial Pricing

Implement

$$u = e^{\sigma\sqrt{\Delta t}},$$

$$d = e^{-\sigma\sqrt{\Delta t}},$$

and

$$p = \frac{e^{r\Delta t} - d}{u - d}.$$

European option prices should then be calculated using backward induction.

15.7.2 Binomial Convergence

Perform calculations for increasing numbers of time steps,

$$N = 10,\ 25,\ 50,\ 100,\ 250,\ 500,\ 1000.$$

Define

$$E_N = \left| C^{(N)}_{\text{Binomial}} - C_{\text{BS}} \right|.$$

The convergence of

$$E_N$$

toward zero should be demonstrated graphically.

15.7.3 Monte Carlo Pricing

Under risk-neutral dynamics,

$$S_T = S_0 \exp\left[\left(r - \frac{1}{2}\sigma^2\right) T + \sigma\sqrt{T}\, Z\right].$$

The Monte Carlo estimator is

$$C_{\text{MC}} = e^{-rT} \frac{1}{M} \sum_{j=1}^{M} \max\left(S_T^{(j)} - K, 0\right).$$

15.7.4 Monte Carlo Convergence

Repeat the calculation for increasing numbers of simulated paths.

For every calculation report

$$C_{\text{MC}},$$

the Monte Carlo standard error, and an appropriate confidence interval.

The analytical Black–Scholes price should be shown on the same convergence plot.

15.7.5 Final Numerical Comparison

The September numerical work should culminate in

$$\boxed{C_{\text{BS}} \approx C_{\text{Binomial}} \approx C_{\text{MC}}.}$$

This comparison should not be restricted to a single option.

Test combinations representing

$$\text{ITM}, \qquad \text{ATM}, \qquad \text{OTM},$$

and

$$\text{short}, \qquad \text{medium}, \qquad \text{long maturity}.$$

15.7.6 Week 4 Expected Output

`binomial.py`
`monte_carlo.py`
`september_validation.ipynb`
Binomial Convergence Plot
Monte Carlo Convergence Plot
Final Numerical Comparison Table.

15.8 September Month-End Checkpoint

At the end of September, the following components should be complete.

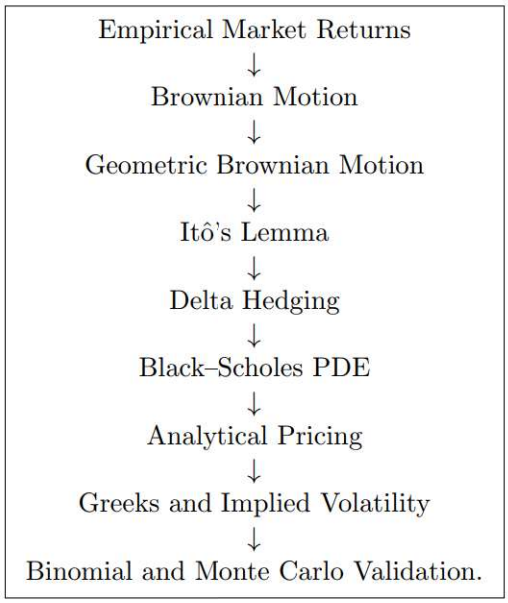
Component	**Target**	**Status**
Brownian-motion simulation	Week 1	☐
GBM simulation	Week 1	☐
NIFTY versus GBM comparison	Week 1	☐
Itô's lemma and BS derivation	Week 2	☐
Call and put pricing code	Week 2	☐
Put–call parity validation	Week 2	☐
Delta, Gamma and Vega	Week 3	☐
Implied-volatility solver	Week 3	☐
Implied-volatility recovery test	Week 3	☐
Binomial pricing	Week 4	☐
Monte Carlo pricing	Week 4	☐
BS–Binomial–MC validation	Week 4	☐

15.9 September Progress Chart

Period	**Principal Focus**	**Expected Progress**
Week 1	Brownian motion, GBM simulation and comparison with empirical NIFTY return behaviour	25%
Week 2	Itô's lemma, delta hedging, Black–Scholes PDE and analytical option pricing	50%
Week 3	Greeks, sensitivity analysis and implied-volatility inversion	75%
Week 4	Binomial and Monte Carlo validation of the analytical pricing framework	100%

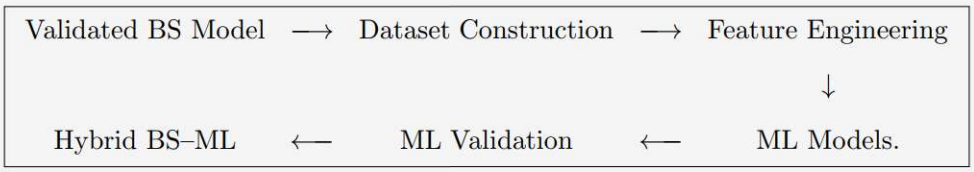
15.10 Expected Position at the End of September

By the end of the month, the complete progression should be



Project Milestone

September should conclude with a mathematically understood and independently validated Black–Scholes computational framework.
The next stage will then begin from a firm baseline:



This will form the basis for the October phase of the project.

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